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(71)Name of Applicant :

1)Birla Vishvakarma Mahavidyalaya Engineering College

Address of Applicant :Mota Bazaar, Vallabh Vidyanagar, Anand, Gujarat 388120, India. -----

2)Yash Roghelia**3)Yash Sojitra****4)Anshul Majmudar****5)Ayush Parmar****6)Prof. Rajiv B. Gandhi****7)Dr. Patel Vinaykumar Jashbhai**

Name of Applicant : NA

Address of Applicant : NA

(72)Name of Inventor :

1)Yash Roghelia

Address of Applicant :38 Rameshwar Park Society, Opp. Madhav Gurukul Bakrol 388315, Gujarat, India. -----

2)Yash Sojitra

Address of Applicant :C-117, Kailash Apartment, Near Chamundanagar L.H. Road, Surat 395006, Gujarat, India. -----

3)Anshul Majmudar

Address of Applicant :19/221, Adarshnagar, Naranpura, Ahmedabad 380013, Gujarat, India. -----

4)Ayush Parmar

Address of Applicant :5,Hari Housing society ,Opp. Lavkush Nagari, Vadsar Road, Vadodara 390010, Gujarat, India. --

5)Prof. Rajiv B. Gandhi

Address of Applicant :7, Avdhoot Society, Nr. Jalaram Society, Nana Bazar, Vallabh Vidyanagar – 388120 Ta/Dist. Anand, Gujarat, India. -----

6)Dr. Patel Vinaykumar Jashbhai

Address of Applicant :C107, Mechanical Engineering Department, Birla Vishvakarma Mahavidyalaya, Vallabhvidyanagar-388120, Dist: Anand, Gujarat, India. -----

(57) Abstract :

An autonomous robot for real-time pathfinding and obstacle avoidance is described herein. The robot comprises pair of wheels (10), two-layer PCB assembly and metal-free moment caster wheels (40). The top layer (20) of PCB assembly further includes ToF sensor (201) positioned in front and two sides of the top layer, 9-axis sensor (202) combining accelerometer, gyroscope, and magnetometer, microcontroller (203), a 8-channel multiplexer (204), and an OLED display (205). The bottom layer (30) of PCB assembly further includes pair of DC encoder motor (301), a motor driver (302), and LED (303). The present invention operates autonomously by combining advanced hardware and algorithms to navigate and solve mazes efficiently. The seamless integration of sensors, power management, and efficient algorithms enables the robot to adapt dynamically to its environment, making it a powerful tool for educational, research, and commercial applications.

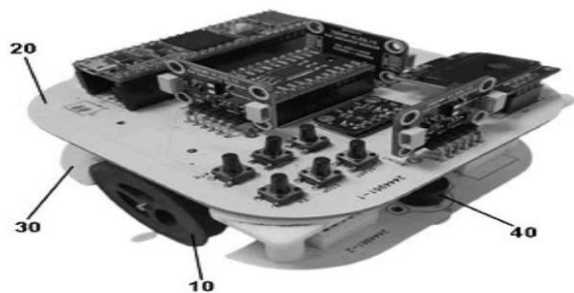


Fig. 1

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